

Thursday Warm-up: Electric Fields in Matter

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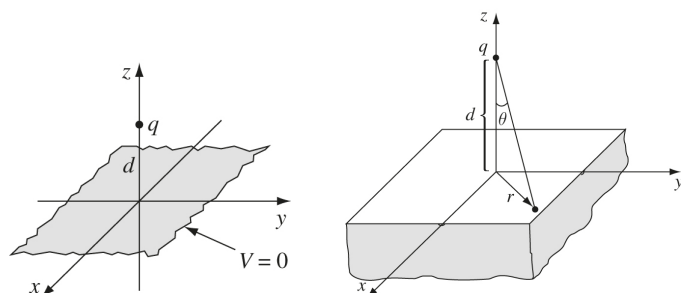


Figure 1: (Left) A free charge above a conductor filling space with $z < 0$. (Right) A free charge above a linear dielectric filling space with $z < 0$.

1 Memory Bank

- **Relative voltage:**

$$V = - \int_a^b \mathbf{E} \cdot d\mathbf{l} \quad (1)$$

- **The electric displacement:**

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad (2)$$

- **Gauss' Law for electric displacement:**

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f,enc} \quad (3)$$

- **Linear dielectric materials:**

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E} \quad (4)$$

$$\mathbf{D} = \epsilon_0 (1 + \chi_e) \mathbf{E} \quad (5)$$

$$\mathbf{D} = \epsilon \mathbf{E} \quad (6)$$

$$\epsilon_r = \epsilon / \epsilon_0 = 1 + \chi_e \quad (7)$$

2 Linear Dielectric Materials

1. Consider Fig. 1 (left). Using the Method of Images, calculate $V(x, y, z)$ and $\mathbf{E}(x, y, z)$ for $z \geq 0$.

2. Consider Fig. 1 (left). (a) Calculate the induced surface charge distribution. (b) Calculate the total induced charge.

3. Consider Fig. 1 (right). The entire region for $z < 0$ is filled with uniform linear dielectric material with susceptibility χ_e . Calculate the force on a point charge q a distance d above the origin.

- First, write an expression for the bound surface charge in terms of E_z , the z-component of the total field just inside the dielectric.

- (a) Write the z-component of the field contribution from q at a point r on the surface. (b) Using Gauss' law, write the field contribution from the surface bound charge.

- (a) Solve for the surface bound charge. (b) Calculate the *total* bound charge, q_b .

- What is the force between q and q_b ? How does this compare to the force between q and the conductor?